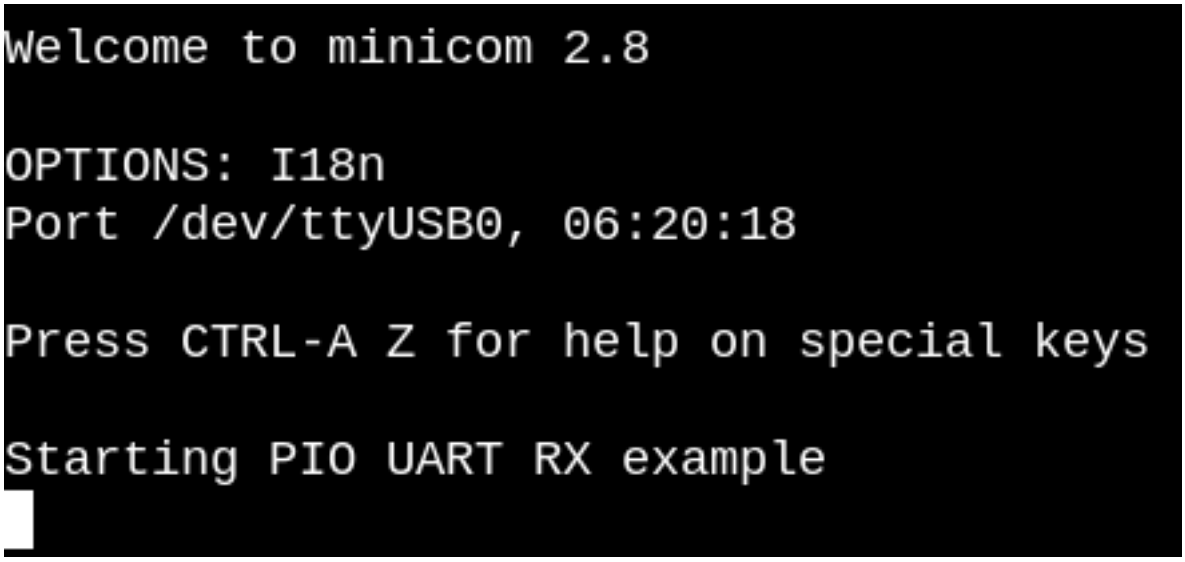
Notes Receiver changes 082126

```
#define CRC32_INIT                ((uint32_t)-1l)
#define DATA_TO_CHECK_LEN          9
#define CRC32_LEN                  4
#define TOTAL_LEN                  (DATA_TO_CHECK_LEN + CRC32_LEN)
static uint8_t src[TOTAL_LEN] = { 0x31, 0x32, 0x33, 0x34, 0x35, 0x36, 0x37, 0x38, 0x39, 0x00,
0x00, 0x00, 0x00 };

static uint8_t src1[TOTAL_LEN] = { 0x39, 0x38, 0x37, 0x36, 0x35, 0x34, 0x33, 0x32, 0x31, 0x00,
0x00, 0x00, 0x00 };
static uint8_t srcR[TOTAL_LEN];
// This uses a standard polynomial with the alternate 'reversed' shift direction.
// It is possible to use a non-reversed algorithm here but the DMA sniff set-up
// below would need to be modified to remain consistent and allow the check to pass.

static uint32_t soft_crc32_block(uint32_t crc, uint8_t *bytp, uint32_t length) {
   while(length--) {
      uint32_t byte32 = (uint32_t)*bytp++;

      for (uint8_t bit = 8; bit; bit--, byte32 >>= 1) {
         crc = (crc >> 1) ^ (((crc ^ byte32) & 1ul) ? 0xEDB88320ul : 0ul);
      }
   }
   return crc;
}



   while (true) {
         if ( invert == false)
      {
            // Invert the input logic hardware-wide (if using as an input)
         gpio_set_inover(PIO_RX_PIN, GPIO_OVERRIDE_INVERT);

            invert = true;
         }

      while (c = uart_rx_program_getc(pio, sm) != 0x2c)
         {
         }

         for(i=0;i < TOTAL_LEN;i++)
         {
            c = uart_rx_program_getc(pio, sm);

            putchar(c);


            srcR[i] = c;
         }

         crc_res = soft_crc32_block(CRC32_INIT, srcR, DATA_TO_CHECK_LEN);
```

```
    printf("0x%x\n",crc_res);

       //9-10xfea0fdfe      1-9      0x340bc6d9
       //                fefda0fe                d9c60b34


}
```



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